

RHO: Robust Holistic OSM-Based Metric Cross-View Geo-Localization

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Abstract

*Metric Cross-View Geo-Localization (MCVGL) aims to estimate the 3-DoF camera pose (position and heading) by matching ground and satellite images. In this work, instead of pinhole and satellite images, we study robust MCVGL using holistic panoramas and OpenStreetMap (OSM). To this end, we establish a large-scale MCVGL benchmark dataset, **CV-RHO**, with over 2.7M images under different weather and lighting conditions, as well as sensor noise. Furthermore, we propose a model termed **RHO** with a two-branch **Pin-Pan** architecture for accurate visual localization. A **Split-Undistort-Merge (SUM)** module is introduced to address the panoramic distortion, and a **Position-Orientation Fusion (POF)** mechanism is designed to enhance the localization accuracy. Extensive experiments prove the value of our CV-RHO dataset and the effectiveness of the RHO model, with a significant performance gain up to 20% compared with the state-of-the-art baselines. The dataset, model, and code will be made publicly available.*

1. Introduction

Cross-View Geo-Localization (CVGL) is one of the fundamental tasks in the computer vision domain. It can be further refined into two sub-directions: Large-Area Cross-View Geo-Localization (LCVGL) [4, 9, 58, 72] and Metric Cross-View Geo-Localization (MCVGL) [38, 39, 52, 53]. LCVGL starts from a large (e.g., city-scale) search region and finds a rough estimate of the camera position. It typically uses an image retrieval approach and therefore does not predict orientation or reach high metric accuracy. MCVGL, on the other hand, starts from a noisy GPS prior (e.g., $\pm 50\text{m}$ error [16]) and determines the position and orientation with higher accuracy at meter level by matching ground and satellite images, which is more challenging and valuable in real-world applications, e.g., autonomous driving [8, 50] and remote sensing [56, 65].

In this work, we study robust MCVGL using panoramas and OpenStreetMap (OSM). The motivation is threefold.

(1) First, the majority of existing studies in the community

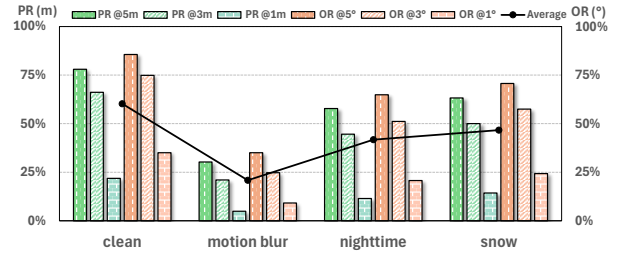


Figure 1. OrienterNet [32] fails under noisy conditions, including motion blur, nighttime, and snowy weather. PR and OR stand for Position Recall and Orientation Recall at three thresholds.

focus on addressing the MCVGL problem under ideal conditions (i.e., sunny with sufficient light). This assumption doesn't always hold. The effects of varying weather and lighting conditions, as well as sensor noise, on MCVGL remain largely underexplored. Fig. 1 showcases that models trained on the dataset under ideal conditions fail at predicting camera poses under noisy conditions with a dramatic performance degradation. (2) Second, in contrast to pinhole images, panoramic images offer richer visual information [14, 66, 67], which facilitates more accurate estimation of the camera's position and orientation. (3) In addition to enhancing robustness and utilizing holistic panoramic representations, incorporating OSM also plays an important role in addressing the MCVGL task. Since satellite imagery is updated infrequently, MCVGL fails when attempting to match newly captured ground images with outdated satellite images. Another advantage of utilizing OSM over satellite imagery lies in its significantly lower storage requirements. While satellite imagery occupies approximately 75MB of storage per 1km^2 , OSM data requires only about one-fifteenth of the space with only 4.8MB [32].

To evaluate the robustness of MCVGL, we create a large-scale dataset, **CV-RHO**, with over 2.7M images under varying conditions and perturbations, including rain, snow, fog, and nighttime scenarios, as well as variations such as overexposure, underexposure, and motion blur, as illustrated in Fig. 2. It's worth noting that all generated images are structurally consistent with the original image.

CV-RHO is the first large-scale benchmark dataset aiming at studying the robustness of OSM-based Metric Cross-View Geo-Localization, providing a promising step towards the robustness and generalization of the CVGL task. Furthermore, we propose a model termed **RHO** with a two-branch **Pin-Pan** architecture. The panoramic branch takes the 360°-FoV images as input and predicts the probability distribution of the camera positions. To handle the panoramic distortion due to the equirectangular projection, a Split-Undistort-Merge (**SUM**) module is introduced and integrated into the panoramic branch. SUM firstly splits one panorama into three 120°-FoV pinhole images and then undistorts the panoramic deformation. All three pinholes go through the image encoder and the Bird's-Eye-View (BEV) projection, resulting in three BEV neural maps, which are finally merged in the BEV space. The fused BEV neural map is utilized to align with the OSM data in order to estimate the distribution of camera positions. This matching results in a three-dimensional (*i.e.*, H, W, Angles) distribution volume. The pinhole branch, on the other hand, estimates the camera heading using the pinhole image derived from the panorama. A Position-Orientation Fusion (**POF**) mechanism is designed to fuse the panoramic and pinhole distribution volumes. The panoramic volume is aggregated along its orientational dimension to refine the pinhole volume, while the pinhole volume is aggregated along its spatial dimensions to enrich the panoramic volume. Extensive experiments on the CV-RHO dataset verify the effectiveness of our proposed RHO model with a significant performance boost of up to 20% under different conditions, outperforming other state-of-the-art baselines on the OSM-based MCVGL task. To summarize, **our contributions** are listed as follows:

- We create the first large-scale OSM-based MCVGL dataset with over 2.7M images, **CV-RHO**, to study the robustness of MCVGL under different weather and lighting conditions, as well as sensor noise, including rain, snow, fog, night, overexposure, underexposure, and motion blur.
- We propose a novel model termed **RHO** with a two-branch **Pin-Pan** architecture. The panoramic branch aims to predict the camera position, while the pinhole branch is for the orientation estimation.
- We introduce a **Split-Undistort-Merge** module to handle the panoramic distortion due to the equirectangular projection.
- We design a **Position-Orientation Fusion** mechanism to mutually enhance the probability distribution of camera position and heading.
- Extensive experiments on the CV-RHO dataset prove the value of the CV-RHO and the effectiveness of the RHO model with a significant performance gain up to 20% over the baseline.

2. Related Work

OSM-Based Cross-View Geo-Localization. Satellite-based CVGL [8, 15, 18, 20, 40, 44, 46, 50, 51, 60] refers to matching the ground and satellite images to predict the position and orientation of the current viewpoint. In contrast to satellite-based CVGL, OSM-based CVGL leverages the OpenStreetMap (OSM) instead of the satellite images during the matching process. OrienterNet [32], as the seminal work, first introduces the concept of OSM-based CVGL. OMSLoc [24] proposes a geometry-guided BEV transform with the foundation model's depth prior, and an auxiliary semantic alignment task that enhances the scene understanding ability of the model. Similar to OMSLoc, DP-Loc [17] leverages the prior depth for boosting the localization accuracy. MapLocNet [48] applies a coarse-to-fine feature registration mechanism to reduce the inference latency. U-BEV [3] presents a lightweight architecture that is geometrically constrained and leverages the height of the points from the ground instead of their depth from the camera. SegLocNet [69] proposes a multimodal localization network that achieves highly accurate localization in GPS-denied areas by incorporating the BEV segmentation and 2D navigation maps. DiffVL [10] integrates diffusion models to denoise noisy GPS trajectories for OSM-based CVGL, achieving better localization accuracy. In this work, we delve into the OSM-based CVGL since OSM is updated more frequently than satellite imagery and is storage-friendly.

Robust Cross-View Geo-Localization. Most of the existing works [5, 22, 27, 33, 34, 41–43, 55, 63] assume that the cross-view matching is under ideal conditions, which doesn't always hold in the real world. Robust CVGL is underexplored. SSPT [6] develops a multi-modal data augmentation strategy for Unmanned Aerial Vehicle (UAV) image datasets, broadening the diversity and complexity of the training data. Weln [28] enhances the model robustness by expanding the University-1652 [68] dataset with nine different weather conditions added. A benchmark dataset [61] for evaluating the robustness of CVGL models is proposed by introducing 16 common types of data corruption. MCGF [7] introduces a multi-weather CVGL framework designed to dynamically adapt to unseen weather conditions. ConGeo [29] enhances robustness and consistency in feature representations to improve a model's invariance to orientation and its resilience to FoV variations. CVD [21] explicitly disentangles content and viewpoint factors for robust CVGL. [12] presents a dual-branch transformer framework with gradient-aware weighting feature alignment for robust CVGL. In addition to a multi-dimensional feature matching mechanism, [45] introduces a center-focused attention mechanism to improve the matching accuracy and robustness. However, there are no benchmarks and methods targeting the OSM-based MCVGL, an emerging and highly promising frontier for visual localization. To bridge this

Table 1. Data distribution of the CV-RHO dataset.

Country	City	360°-FoV Pan.	120°-FoV Pin.	OSM
USA	Chicago	12.4K	37.2K	12.4K
	Detroit	39.4K	118.1K	39.4K
	San Francisco	15.5K	46.5K	15.5K
	Washington	12.5K	37.5K	12.5K
France	Montrouge	9.5K	28.5K	9.5K
	Toulouse	12.4K	37.1K	12.4K
Germany	Berlin	12.4K	37.1K	12.4K
Total		114.0K	342.1K	114.0K
8 variations in total		912.3K	2.7M	912.3K

gap, we propose the first large-scale OSM-based MCVGL dataset with over 2.7M images, CV-RHO, as well as training a robust model, RHO, against diverse perturbations.

Panoramic Cross-View Geo-Localization. Narrow-FoV ground images [30, 31, 35–37, 54] provide limited visual information to generate BEV features on the CVGL task, resulting in low accuracy in visual localization. Panoramas [13, 49, 59, 70], on the other hand, enable a holistic understanding of the environments and generate 360°-FoV BEV features, achieving more accurate cross-view matching. SIRNet [26] decomposes the complex learning process into several refinement steps while adapting the refinement steps specifically for each panoramic input. GeoDTR [62] explicitly disentangles geometric information from raw features and learns the spatial correlations among visual features from aerial and panoramic ground pairs. GeoDTR+ [64], built upon GeoDTR, models the spatial configuration of both panoramic and aerial images through the geometric layout extraction mechanism. TriViTs [1] is designed to leverage both spatially aligned and original cross-view images. DSTG [23] enhances panoramic image representation through a “pretraining + conventional training” optimization strategy. However, these methods only apply panoramas to the satellite-based CVGL task. Leveraging the information-rich panoramas on the OSM-based CVGL task remains unexplored. In this work, we apply 360°-FoV images to the OSM-based MCVGL task to improve localization accuracy.

3. Methodology

3.1. OSM-Based MCVGL

Definition. For a single image $\mathbf{I}$ with coarse position prior $\mathbf{x}_{prior}$ and known camera calibration, we aim to query the precise camera pose by matching the ground image and OSM. As OSM consists mainly of 2D geospatial data, it is reasonable to reduce the normal 6-DoF pose estimation to the 3-DoF pose $\xi = (u, v, \theta)$, where (u, v) stands for position and $\theta \in (-180^\circ, 180^\circ]$ is for orientation.

Table 2. Comparison between our CV-RHO dataset and other widely-used CVGL datasets.

Dataset	Pin.	Pan.	Satellite	OSM	Weather noise	Sensor noise
CVUSA [47]	×	✓	✓	×	×	×
CVACT [25]	×	✓	✓	×	×	×
VIGOR [71]	×	✓	✓	×	×	×
CVGlobal [57]	×	✓	✓	✓	×	×
KITTI [11]	✓	×	✓	✓	×	×
MGL [32]	✓	×	×	✓	×	×
CV-RHO (ours)	✓	✓	✓	✓	✓	✓

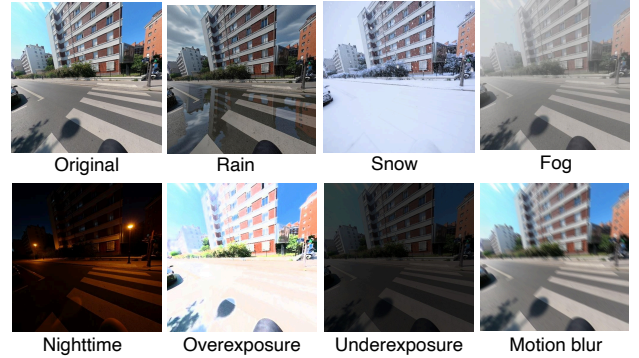


Figure 2. Data samples of our CV-RHO dataset.

Conventions. We assume the direction of gravity is known, which can be assessed via an inertial unit embedded in camera devices. We consider a topocentric coordinate system, with the x-y-z axes corresponding to the Earth-North-vertical directions. For the input image $\mathbf{I}$, the roll and pitch angles are rectified to zero using the known gravity direction, and the principal axis is ensured to be horizontal. The OSM is queried as a square area centered around $\mathbf{x}_{prior}$, whose coarse extent decides how large the extracted map is.

3.2. CV-RHO Dataset Collection

We collect all panoramas and the corresponding metadata using the Mapillary platform. Given the metadata, we also collect the corresponding OSM tiles. Regarding varying weather and lighting conditions, FLUX.1 Kontext [19] is applied for image generation under rain, snow, fog, and night, which are representative of common scenarios in MCVGL real-world applications. The whole image generation process takes 30720 A100 GPU hours. The prompts and parameters used for generating each weather condition are provided in our supplementary material. For perturbations like motion blur, overexposure, and underexposure, OpenCV [2] is used to introduce noise effects into the original images. These three variations involve changes in brightness and exposure duration, which effectively replicate real-world conditions. Fig. 2 presents data samples of the CV-RHO dataset under varying conditions. It is worth noting that all generated images are structurally consistent with the original image. Table 1 and 2 showcase the data

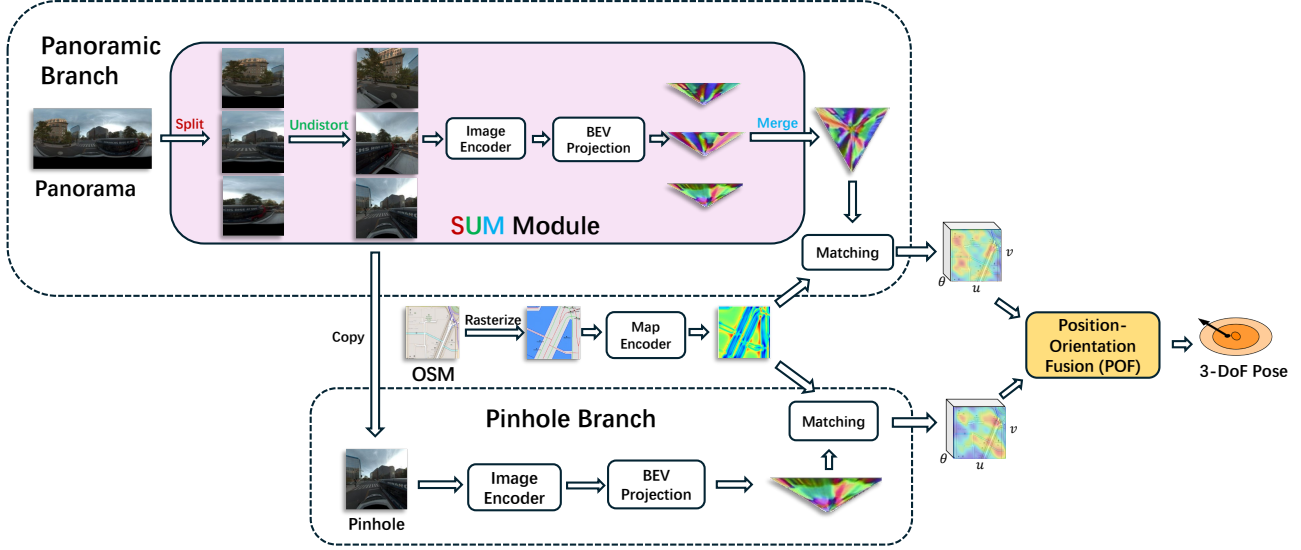


Figure 3. Overview of the RHO model in a two-branch Pin-Pan architecture. Given the current panorama as input, the panoramic branch with Split-Undistort-Merg (SUM) module produces the 360°-FoV feature, while the pinhole branch focuses on local dense feature in 120°FoV. Both generated features are matched with the OSM feature map by using the proposed Position-Orientation Fusion (PoF).

distribution of the CV-RHO dataset and the comparison with other widely-used CVGL datasets, respectively. We split the dataset into train and test sets with the ratio of 8:2. The validation set uses the same set as test set.

To assess the quality of the generated data, we additionally collect a test set with 5K images and evaluate the RHO model using the additional test set in a zero-shot manner. This Sim2Real test set contains photos captured under real-world weather perturbations. The results are listed in Table 6 and will be discussed in Section 4.

3.3. RHO Model

RHO Model Structure. Fig. 3 presents the overview of our RHO model. RHO adopts a two-branch Pin-Pan architecture. The Shannon Entropy of an image is defined as:

$$H = - \sum_i p_i \log_2 p_i, \quad (1)$$

where p_i stands for the occurring probability of the i -th pixel. Compared to pinhole images, a panorama provides richer visual information for the estimation of the camera position due to the larger FoV, resulting in larger Shannon Entropy. However, referring to the prediction of the orientation, the panorama is hard to bring additional performance gain since the Shannon Entropy remains the same when the camera rotates by an angle. In this case, the pinhole image showcases its potential in predicting the orientation, while the Shannon Entropy differs after the camera’s rotation.

The panoramic branch processes a panoramic image to produce a BEV feature map covering a full 360° FoV, while

the pinhole branch takes a pinhole image as input and generates a BEV feature map with a 120° FoV. The OSM tile is first rasterized and then fed into a map encoder, resulting in an OSM feature map. The two BEV feature maps are then respectively matched with the OSM feature map, outputting two probability volumes. Each volume has three dimensions, namely u , v and θ representing width, height and the number of rotation angles, respectively. These two volumes are fused by the Position-Orientation Fusion (POF) module, and the RHO model finally estimates the 3-DoF camera pose.

SUM Module. A panoramic ground image is split into 3 views. Each view is a 120°-FoV pinhole image representing the yaw angle ranging in $[0^\circ, 120^\circ)$, $[120^\circ, 240^\circ)$, and $[240^\circ, 360^\circ)$, respectively. After the splitting, all pinhole images are then undistorted to remove the panoramic deformation. Each pinhole image $\mathbf{I}_i, i \in \{1, 2, 3\}$ is encoded via Φ_{image} and Φ_{bev} forming a BEV feature map $\mathbf{X}_i, i \in \{1, 2, 3\}$. All $\mathbf{X}_i$ are then merged, resulting in a panoramic BEV feature map $\mathbf{X}_{pan}$.

Map Encoding. The 2D map with areas, lines, and points classes is first rasterized as a 3-channel image. Each semantic class is then mapped to a unique semantic vector. We employ the Φ_{map} to encode the rasterized map into a neural map $\mathbf{M}$, which will be utilized in the map-BEV matching.

POF Module. By exhaustively matching the neural map $\mathbf{M}$ and panoramic BEV feature map $\mathbf{X}_{pan}$, we yield a score volume S_{pano} . Meanwhile, we calculate the score volume S_1 by matching the neural map $\mathbf{M}$ and the pinhole BEV feature map $\mathbf{X}_1$ of the front view, representing the yaw angle

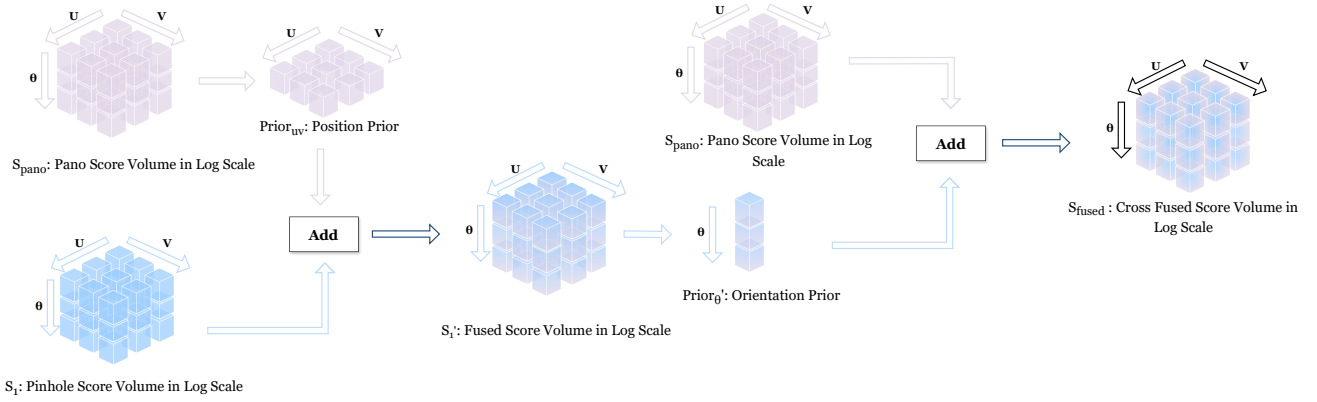


Figure 4. Position-Orientation Fusion module of the RHO model. Given both 360°-FoV and 120°-FoV features, POF combines the pinhole volume S_1 with $Prior_{uv}$, and then the panoramic volume S_{pano} with $Prior_{\theta}$, respectively.

ranging in $[0^\circ, 120^\circ]$. S_{pano} contains accurate prediction of position, while S_1 focuses more on the orientation. In order to leverage the advantages of both S_{pano} and S_1 , we propose a POF module, a two-stage fusion process shown in Fig. 4. In the first stage, for location $\mathbf{x} = (u, v)$, and camera pose $\xi = (\mathbf{x}, \theta)$, we calculate a robust 2D spatial prior by marginalizing the panorama log probability volume over orientation dimension θ using $LogSumExp(LSE)$. The 2D prior highlights likely locations regardless orientation.

$$\log P_{pano}(\xi) = LogSoftmax(S_{pano}(\xi)) \quad (2)$$

$$\log P_{uv,prior}(\mathbf{x}) = LSE_{\theta}(\log P_{pano}(\xi)) \quad (3)$$

In the second stage, the robust spatial prior $\log P_{uv,prior}(\mathbf{x})$ guides the position estimation of the pinhole view, which is more sensitive to orientation. First, the pinhole log-probabilities are re-weighted by the spatial prior. A sharp orientation prior is obtained by normalizing these corrected scores and marginalizing over the spatial dimensions $\mathbf{x}$.

$$\log P_{v1}(\xi) = LogSoftmax(S_{v1}(\xi)) \quad (4)$$

$$S'_{v1}(\xi) = (1 - \alpha) \cdot \log P_{v1}(\xi) + \alpha \cdot \log P_{uv,prior}(\mathbf{x}) \quad (5)$$

$$\log P'_{v1}(\xi) = LogSoftmax(S'_{v1}(\xi)) \quad (6)$$

$$\log P_{\theta,prior}(\theta) = LSE_{\mathbf{x}}(\log P'_{v1}(\xi)) \quad (7)$$

In the final stage, the robust panorama probability volume S_{pano} is fused with the precise orientation prior $\log P_{\theta,prior}(\theta)$ as the final fused score volume S_{fused} .

$$S_{fused}(\xi) = (1 - \beta) \cdot \log P_{pano}(\xi) + \beta \cdot \log P_{\theta,prior}(\theta) \quad (8)$$

The final score volume S_{fused} is then used for determining the final pose estimated by maximizing likelihood $\xi^* = \argmax_{\xi} P(\xi | \mathbf{I}, map, \xi_{prior})$.

Table 3. Results on clean CV-RHO dataset. Position Recall (PR) and Orientation Recall (OR) are in percentages.

Approach	FoV	PR @ X _m			OR @ X°		
		1m	3m	5m	1°	3°	5°
OrienterNet [32]	90°	18.02	58.37	71.04	27.72	63.86	77.50
OrienterNet [32]	120°	21.83	66.16	78.03	35.02	74.89	85.62
OrienterNet [32]	360°	3.79	19.35	28.78	10.29	28.43	36.87
RHO (ours)	360°	24.59	73.55	84.36	43.46	83.61	90.44

4. Experiments

4.1. Implementation Details

We conduct experiments on our large-scale CV-RHO dataset. We use 12 NVIDIA A100 GPUs to train the RHO model. The number of rotations is 64 during training and 256 in evaluation. The batch size and learning rate are set to 36 and $2e-5$, respectively. We choose ReduceLROnPlateau as the learning rate scheduler to manage the learning rate and Adam optimizer for training. With the settings above, the model achieves the best result between epochs 2 and 4, step 20k to 40k approximately.

4.2. Quantitative Results

Results on Clean CV-RHO Table 3 showcases the results on the clean CV-RHO dataset, namely using clean, daytime ground images for training. When trained with 120°-FoV pinhole images, OrienterNet achieves better recall than when trained with 90°-FoV images on CV-RHO. This shows that a broader FoV boosts the model performance. However, the 360°-FoV OrienterNet has the worst results due to the lack of the ability to handle panoramic distortion. Our model RHO outperforms other baselines with significant performance gains of 2.76% / 7.39% / 6.33% on PR@1/3/5m and 8.44% / 8.72% / 4.82% on

Table 4. Results on CV-RHO dataset under varying conditions and noises. Position Recall (PR) and Orientation Recall (OR) are in percentages. NT: nighttime; OE: over-exposure; UE: under-exposure; MB: motion blur; ANV: all noisy variations; AV: all variations; Average Degradation is overall performance drop.

Method	Variation		PR @ Xm ↑			OR @ X° ↑		
	Train	Eval	1m	3m	5m	1°	3°	5°
OrienterNet [32]	Clean	Clean	21.83	66.16	78.03	35.02	74.89	85.62
	Clean	Rain	13.98	52.98	68.12	25.51	60.49	74.68
	Clean	NT	11.44	44.55	57.77	20.71	51.13	64.84
	Clean	Fog	16.26	54.52	67.12	26.81	62.27	75.02
	Clean	Snow	14.29	50.02	63.23	24.30	57.52	70.68
	Clean	OE	16.36	57.57	70.86	28.90	66.00	78.55
	Clean	UE	19.74	63.25	75.65	32.88	71.90	83.35
	Clean	MB	4.90	20.98	30.30	9.15	24.72	35.05
	Clean	ANV	12.41	46.43	60.03	23.51	55.06	67.44
	Clean	AV	13.15	48.46	62.17	24.56	57.25	69.63
Average Degradation			-8.22	-17.41	-16.34	-10.98	-18.63	-16.82
RHO (ours)	Clean	Clean	24.59	73.55	84.36	43.46	83.61	90.44
	Clean	Rain	20.91	68.48	80.77	37.58	78.25	87.19
	Clean	NT	17.46	59.83	72.48	30.74	68.37	79.51
	Clean	Fog	20.27	64.42	75.93	35.23	72.63	82.23
	Clean	Snow	18.95	62.33	74.14	32.35	70.45	80.73
	Clean	OE	20.23	67.73	79.46	38.17	78.12	86.51
	Clean	UE	23.49	71.95	83.17	41.93	82.25	89.37
	Clean	MB	10.15	37.50	49.69	18.15	45.06	57.82
	Clean	ANV	17.67	60.08	73.01	32.99	70.51	80.52
	Clean	AV	18.48	61.51	74.38	34.47	72.01	81.77
Average Degradation			-5.97	-12.01	-10.69	-9.95	-12.76	-9.81
OrienterNet [32]	Rain	Rain	20.48	64.36	76.57	33.89	73.41	84.49
	NT	NT	19.26	60.76	73.04	31.75	70.06	81.42
	Fog	Fog	20.32	63.39	75.53	33.26	72.42	83.43
	Snow	Snow	20.29	63.56	75.68	33.57	72.54	83.72
	OE	OE	20.32	63.81	76.15	33.09	72.58	83.88
	UE	UE	21.36	65.16	77.41	34.80	74.31	85.02
	MB	MB	19.78	62.29	74.56	32.68	71.33	82.63
	ANV	ANV	18.13	60.36	73.70	32.89	71.38	82.24
	AV	AV	18.14	60.72	74.21	32.83	71.48	82.34
Average Degradation			-2.04	-3.45	-2.82	-1.82	-2.72	-2.38
RHO (ours)	Rain	Rain	24.8	73.91	84.55	42.28	83.35	90.63
	NT	NT	24.46	72.04	83.01	41.23	81.53	89.44
	Fog	Fog	25.34	73.12	83.93	41.89	82.72	90.35
	Snow	Snow	24.75	72.68	83.56	43.09	82.72	89.82
	OE	OE	24.94	73.20	83.80	41.68	82.05	89.83
	UE	UE	25.91	74.42	84.72	43.23	83.61	90.67
	MB	MB	25.39	73.45	84.07	42.13	82.63	90.15
	ANV	ANV	23.02	72.11	83.58	42.52	82.43	90.00
	AV	AV	22.98	71.58	83.32	43.22	82.50	89.91
Average Degradation			+0.03	-0.60	-0.52	-1.10	-0.99	-0.35

Table 5. Results of cross-region evaluation. OrienterNet and our model RHO are both trained on the CV-RHO under various conditions and evaluated on the new city dataset. Position Recall (PR) and Orientation Recall (OR) are in percentages. MV: Mount Vernon. NT: nighttime; OE: over-exposure; UE: under-exposure; MB: motion blur; ANV: all noisy variations; AV: all variations.

Method	Variation	Dataset	PR @ Xm ↑			OR @ X° ↑		
			1m	3m	5m	1°	3°	5°
OrienterNet [32]	Clean	MV	16.36	54.17	62.82	25.62	58.56	69.80
RHO (ours)			26.24	67.38	73.79	38.00	73.35	80.20
OrienterNet [32]			21.83	66.16	78.03	35.02	74.89	85.62
RHO (ours)			24.59	73.55	84.36	43.46	83.61	90.44
OrienterNet [32]	Rain	MV	15.94	53.03	61.75	25.34	57.82	69.40
RHO (ours)			26.37	66.75	73.27	37.79	72.67	78.90
OrienterNet [32]			20.48	64.36	76.57	33.89	73.41	84.49
RHO (ours)			24.80	73.91	84.55	42.28	83.35	90.63
OrienterNet [32]	NT	MV	14.14	48.59	57.83	22.71	54.33	65.67
RHO (ours)			25.10	65.27	71.30	36.13	70.21	77.76
OrienterNet [32]			19.26	60.76	73.04	31.75	70.06	81.42
RHO (ours)			24.46	72.04	83.01	41.23	81.53	89.44
OrienterNet [32]	Fog	MV	15.43	51.55	60.78	24.13	56.54	68.22
RHO (ours)			23.83	65.07	72.49	36.41	71.76	78.15
OrienterNet [32]			20.32	63.39	75.53	33.26	72.42	83.43
RHO (ours)			25.34	73.12	83.93	41.89	82.72	90.35
OrienterNet [32]	Snow	MV	13.71	48.60	57.93	23.74	54.58	66.29
RHO (ours)			23.62	64.21	71.58	37.50	71.35	78.22
OrienterNet [32]			20.29	63.56	75.68	33.57	72.54	83.72
RHO (ours)			24.75	72.68	83.56	43.09	82.72	89.82
OrienterNet [32]	OE	MV	14.92	51.37	60.66	24.60	56.89	68.36
RHO (ours)			24.42	62.78	69.06	35.38	69.35	75.86
OrienterNet [32]			20.32	63.81	76.15	33.09	72.58	83.88
RHO (ours)			24.94	73.20	83.80	41.68	82.05	89.83
OrienterNet [32]	UE	MV	16.33	53.48	62.60	25.85	59.17	70.72
RHO (ours)			25.51	64.78	70.98	35.95	70.85	77.68
OrienterNet [32]			21.36	65.16	77.41	34.80	74.31	85.02
RHO (ours)			25.91	74.42	84.72	43.23	83.61	90.67
OrienterNet [32]	MB	MV	13.42	48.30	57.82	23.28	53.85	66.17
RHO (ours)			24.60	64.31	71.71	35.56	70.41	77.60
OrienterNet [32]			19.78	62.29	74.56	32.68	71.33	82.63
RHO (ours)			25.39	73.45	84.07	42.13	82.63	90.15
OrienterNet [32]	ANV	MV	14.58	49.77	58.69	24.20	55.56	66.81
RHO (ours)			24.96	65.32	71.58	36.53	72.03	78.20
OrienterNet [32]			18.13	60.36	73.70	32.89	71.38	82.24
RHO (ours)			23.02	72.11	83.58	42.52	82.43	90.00
OrienterNet [32]	AV	MV	14.51	50.24	59.29	24.77	56.58	67.31
RHO (ours)			25.10	65.78	72.13	37.65	72.72	78.48
OrienterNet [32]			18.14	60.72	74.21	32.83	71.48	82.34
RHO (ours)			22.98	71.58	83.32	43.22	82.50	89.91

OR@1/3/5°. This performance boost proves that RHO achieves the best results due to the reasonable model design rather than purely leveraging large-FoV images.

Results on Noisy CV-RHO. We first train OrienterNet and RHO using the clean variation of the CV-RHO dataset, then evaluate them on all variations. The results are reported in Table 4. It can be observed that both OrienterNet and RHO have performance degradation in such a setting. However, the performance drop of RHO is lower than OrienterNet on both positional and orientational recalls. It proves that RHO is superior on the OSM-based MCVGL task. We then train and evaluate OrienterNet and RHO using every

single variation. All results are listed in the lower part of Table 4. Both OrienterNet and RHO are able to handle different weather and lighting conditions, as well as sensor noise, after training on noisy CV-RHO. It proves that models trained on our proposed CV-RHO dataset are capable of handling different conditions and perturbations. CV-RHO can be used to benchmark the robust OSM-based MCVGL task. Moreover, RHO consistently outperforms OrienterNet in the noisy setting.

Results of Cross-Region Evaluation. We additionally collect a test set with 10k images in Mount Vernon, USA, for the sake of evaluating the cross-region localization ability of the RHO model. The results are reported in Table 5. It

Table 6. Sim2Real Results. Position Recall (PR) and Orientation Recall (OR) are in percentages. MV: Mount Vernon; OE: over-exposure; UE: under-exposure; MB: motion blur.

Method	Dataset	Variation	City	PR @ Xm ↑			OR @ X° ↑		
				1m	3m	5m	1°	3°	5°
OrienterNet [32]	Sim2Real	Rain	Ogori	4.85	40.20	56.36	18.59	47.47	61.62
		Snow	Detroit	11.16	47.07	57.54	22.59	52.13	63.95
		OE	Hasselt	3.97	29.71	47.84	17.62	47.67	63.04
		UE	Hasselt	2.07	17.10	28.67	10.19	27.29	39.03
		MB	Hasselt	3.87	28.62	45.29	15.99	46.63	62.12
	Cross Region	Rain	MV	15.94	53.03	61.75	25.34	57.82	69.40
		Snow		13.71	48.6	57.93	23.74	54.58	66.29
		OE		14.92	51.37	60.66	24.6	56.89	68.36
		UE		16.33	53.48	62.60	25.85	59.17	70.72
		MB		13.42	48.30	57.82	23.28	53.85	66.17
RHO (ours)	Sim2Real	Rain	Ogori	3.64	54.55	68.48	32.73	69.09	81.82
		Snow	Detroit	16.32	54.03	62.91	33.16	66.63	72.52
		OE	Hasselt	5.18	36.79	62.69	23.83	59.07	76.17
		UE	Hasselt	6.70	42.27	61.86	24.74	60.31	76.29
		MB	Hasselt	3.54	42.93	63.13	27.27	61.11	79.29
	Cross Region	Rain	MV	26.37	66.75	73.27	37.79	72.67	78.90
		Snow		23.62	64.21	71.58	37.50	71.35	78.22
		OE		24.42	62.78	69.06	35.38	69.35	75.86
		UE		25.51	64.78	70.98	35.95	70.85	77.68
		MB		24.60	64.31	71.71	35.56	70.41	77.60

can be observed that RHO consistently outperforms OrienterNet under all condition variations, even though we train RHO using CV-RHO while evaluating the model using the cross-region test set. It proves the excellent cross-region visual localization ability of our proposed RHO.

Results of Sim2Real Evaluation. We additionally collect a test set with 5k real-world images under different conditions to study the Sim2Real gap. All models are trained on CV-RHO while evaluated on both Sim2Real and Cross-Region datasets. Since the Sim2Real dataset is collected in the new cities unseen in CV-RHO, we compare the Sim2Real results with the Cross-Region results instead of the CV-RHO test set results for fairness. The results are listed in Table 6. It can be observed that even though the Sim2Real gap exists, models trained on CV-RHO still achieve high position and orientation recalls, especially on the PR@5m and OR@5° metrics, proving the value of our proposed CV-RHO dataset and the effectiveness of the RHO model.

4.3. Ablation Study

Analysis of BEV feature shape. We analyze various shapes and merge strategies of pinhole BEV features. A 120° FoV pinhole BEV feature is in shape of an isosceles triangle with a 120° vertex angle. By rotating the 3 views anticlockwise 0°, 120°, and 240° respectively, we concatenate them into an equiangular triangle to represent

Table 7. Results of different merge strategies evaluated with the clean data of Washington in our CV-RHO dataset. 120° FoV pinhole features concatenated into an equiangular triangle achieves the best performance in merged features. Pinhole features show their powerful orientation prediction if not merged.

FoV	Merged	Position R @ Xm ↑			Orientation R @ X° ↑		
		1m	3m	5m	1°	3°	5°
120°	✓	14.44	68.14	80.6	18.61	51.47	69.87
90°		3.21	21.82	34.22	16.17	35.79	41.26
60°		0.34	3.04	7.34	0.90	2.55	4.07
40°		0.29	2.82	6.39	10.4	23.63	30.68
120°	✗	12.04	60.37	75.24	31.52	71.21	83.74
90°		10.42	51.03	66.36	24.18	58.58	73.29

the panoramic BEV feature. If a panorama is split into 4 views, each with a 90° FoV, the pinhole feature is in the shape of an isosceles right triangle. After rotation the 4 pinhole BEV feature maps can be merged into a square.

Besides, we split a panorama further into 60° and 40° pinhole views. We use them to represent a 120° pinhole feature by rotating and concatenating them into a rhombus and a trapezium, respectively. However, they do not show a better performance compared with using a pinhole view in the same FoV directly. Results in Table 7 show that an equiangular form outperforms all other merged shapes, and

Table 8. Analysis of different fusion strategies.

Strategy	Position R @ $X_m \uparrow$			Orientation R @ $X^\circ \uparrow$		
	1m	3m	5m	1°	3°	5°
No fusion	14.44	68.14	80.60	18.61	51.47	69.87
$Prior_{uv}$ fusion	11.93	59.47	77.38	30.29	73.13	85.14
$Prior_\theta$ fusion	13.66	68.63	81.76	32.15	78.21	88.69
POF (ours)	14.53	70.33	83.16	37.39	82.96	91.21

with a performance gain up to 8% in position recall. This proves our SUM strategy splitting panoramas into 120° FoV pinhole images is a reasonable approach to represent the panoramic features.

Analysis of fusion strategy. We found that although the merged panoramic outperforms on position recall, the pinhole view is more accurate in predicting orientation angle. The performance gap between merged and non-merged methods achieves a maximum of 20% at orientation recall. This leads us to our fusion strategy, which combines the remarkable positioning ability of panoramic features with the precise orientation prediction of pinhole features.

In Table 8, we explored several possible strategies that could be suitable for scores volume fusion. First, we tried to extract the precise position prior from the panoramic score volume S_{pano} and add it directly to the $View_1$ score volume S_1 . This is an intuitive approach to fuse them together. Second, we try to extract the accurate orientation prior from S_1 and add it to S_{pano} . Results in Table 8 show that taking S_{pano} rather than S_1 as the base score volume to estimate the final pose achieves higher recall both in position and orientation. Taking S_{pano} as the base score volume boosts the orientation recall up to 89% at the threshold 5°. This suggests that the S_{pano} should be the main score volume. However, these two fusion methods involve only rectification on one score volume, but we want S_1 and S_{pano} to influence each other mutually. Therefore, we designed our final fusion strategy to extract the position prior from S_{pano} first, and use it to rectify S_1 . Next, the rectified S_1 produces an accurate orientation prior, and uses it to correct S_{pano} . Each fusion stage has a learnable parameter to manage the weight of priors. Results in Table 8 confirm our cross-fused strategy maximizes the strengths of panoramas and pinholes in terms of positioning and orientation. Our POF module demonstrates orientation recall improvements of up to 20% compared to using S_{pano} with no fusion. Meanwhile, the position prediction preserves its accuracy and yields a moderate performance enhancement.

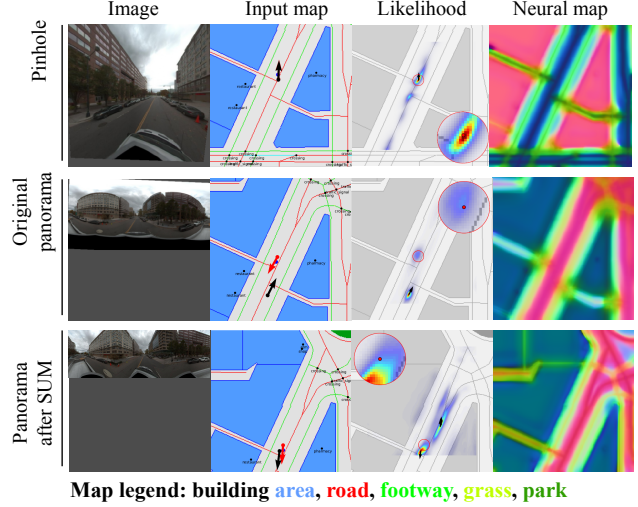


Figure 5. Qualitative analysis of different FoVs. Overlaid on the input maps, the camera pose predictions (black arrow $\rightarrow$) are close to the ground truth (red arrow $\rightarrow$) and more accurate than the noisy GPS (blue dot $\bullet$). We show the predicted orientations at local maxima as small arrows on likelihood maps.

4.4. Qualitative Analysis

In Fig. 5, we visualize some qualitative examples of evaluating OrienterNet with multimodal inputs. The results show that (1) OrienterNet cannot perform an accurate prediction while taking original 360° panoramas directly as the input. (2) Our RHO model using the panorama after processed by SUM module enhances the prediction effectiveness significantly, as it is shown in the input map and likelihood. (3) The neural map leverages building corners and boundaries, sidewalks, road intersections, trees, and other common urban objects more effectively.

5. Conclusion

In this work, we introduced CV-RHO, the first large-scale benchmark for evaluating the robustness of OSM-based MCVGL across diverse weather, lighting, and sensor conditions. With over 2.7M panoramic and pinhole images from multiple cities and eight realistic variations, CV-RHO enables comprehensive assessment beyond clean settings. Building on this dataset, we proposed RHO, a robust metric geo-localization model with a two-branch Pin-Pan architecture. The panoramic branch leverages 360° imagery with our SUM module to correct projection distortion, while the pinhole branch provides precise orientation cues. Our POF mechanism effectively fuses the strengths of both branches. Experiments demonstrate that RHO surpasses existing methods and maintains strong performance under challenging conditions, showing solid generalization to adverse real-world scenarios and unseen cities.

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RHO: Robust Holistic OSM-Based Metric Cross-View Geo-Localization

Supplementary Material

A. More Implementation Details

A batch size of 36 was employed when training exclusively with a single **CV-RHO** variation. In contrast, a larger batch size of 96 was utilized for training involving all noisy variations or the complete set of data variations. We limit the maximal training epoch to 20, and the best model checkpoint is selected with early stopping based on validation loss. The evaluation batch size is set to 3. This value represents the minimal batch size for the RHO model, as the SUM module requires the input of at least three pinhole images for correct operation.

B. Prompt for Data Generation of CV-RHO

To achieve a realistic simulation of real-world scenarios under diverse weather conditions without destroying the original buildings and streets' structure in the image, we carefully design prompts and test parameters listed in Table 9 for generating **CV-RHO** images. All weather variations are generated with seed value 42 and true cfg scale 1.2, in inference steps 50. The guidance scale value differs among the 4 variations.

Utilizing the SOTA and open-source image generation model FLUX.1 Kontext [19], we edit images to produce photorealistic variations under diverse weather conditions. Fig. 7 presents additional samples of generated images. The generated weather effects are various, rather than being static or uniform, which corresponds to the randomness in the real world.

C. Data Collection of CV-RHO

We selected sequences captured after 2020 using panoramic cameras. Fig. 10 shows OSM covered by the selected sequences. Images of each location were split into disjoint

training and validation sets; the ratio between them is approximately 8:2, resulting in **2.16M** training and **540K** validation views. Data collected in Mount Vernon is used for cross-region evaluation. In the **Sim2Real** dataset, the Detroit snow area does not overlap with the area where clean data is collected. Data collected in Hasselt is utilized for sensor noise in **Sim2Real**.

D. Detail of Fusion Strategies

Table 8 shows the performance of different fusion strategies. The **POF** fusion strategy outperforms $Prior_{uv}$ fusion and $Prior_{\theta}$ fusion, which are components of **POF** itself. These two fusion strategies are illustrated in Fig. 8. $Prior_{uv}$ fusion marginalizes out the orientation dimension and extracts the position dimensions probability distribution $Prior_{uv}$ from the panoramic score volume S_{pano} and adds it directly to the score volume S_1 to rectify its position prediction. In contrast, $Prior_{\theta}$ fusion tries to marginalize out the positional dimensions and extract the orientation prior from S_1 and add it to S_{pano} . If no fusion strategy is employed, we use S_{pano} directly to estimate the final pose.

E. More Analysis of BEV Merge Shapes

Results in Table 7 show the impact of different BEV merge shapes on the performance. We visualize all tested BEV merge feature shapes in Fig. 9. In the panoramic branch, we try to merge three 120° FoV BEV feature maps into an equiangular triangle, and four 90° FoV BEV feature maps into a square. In the pinhole branch, we merge two 60° FoV BEV feature maps into a rhombus, and three 40° FoV BEV feature maps into a pentagon.

Table 9. Prompts and Parameters configuration for generating images with Flux.1 Kontext [19].

Variation	Prompt	Hyperparameter
Rain	Change the background into a heavy rainy day, change the sky into gray overcast sky, add visible rain streaks and puddles.	Guidance scale: 7.5
Snow	Change the background into a snowy day, add visible falling snowflakes	Guidance scale: 7.7
Fog	Turn the background into a foggy day, add visible smog on the street	Guidance scale: 7.7
Nighttime	Change the background into evening, illuminate the scene brighter with streetlights.	Guidance scale: 5.0
Over Exposure	-	Brightness factor: 2.5
Under Exposure	-	Brightness factor: 0.25
Motion Blur	-	Kernel size: 10

Table 10. Generalization of OrienterNet and RHO in a new city. Δ represents the difference between two recall values: the model’s recall when tested on 7 existing cities versus its recall when tested on the new city.

Method	Variation	PR @ Xm ↑			OR @ X° ↑		
		1m	3m	5m	1°	3°	5°
Δ_{ONet}	Clean	5.47	11.99	15.21	9.4	16.33	15.82
	Rain	4.54	11.33	14.82	8.55	15.59	15.09
	NT	5.12	12.17	15.21	9.04	15.73	15.75
	Fog	4.89	11.84	14.75	9.13	15.88	15.21
	Snow	6.58	14.96	17.75	9.83	17.96	17.43
	OE	5.4	12.44	15.49	8.49	15.69	15.52
	UE	5.03	11.68	14.81	8.95	15.14	14.3
	MB	6.36	13.99	16.74	9.4	17.48	16.46
	ANV	3.55	10.59	15.01	8.69	15.82	15.43
	AV	3.63	10.48	14.92	8.06	14.9	15.03
	Avg.	5.057	12.147	15.471	8.954	16.052	15.604
Δ_{RHO}	Clean	-1.65	6.17	10.57	5.46	10.26	10.24
	Rain	-1.57	7.16	11.28	4.49	10.68	11.73
	NT	-0.64	6.77	11.71	5.1	11.32	11.68
	Fog	1.51	8.05	11.44	5.48	10.96	12.2
	Snow	1.13	8.47	11.98	5.59	11.37	11.6
	OE	0.52	10.42	14.74	6.3	12.7	13.97
	UE	0.4	9.64	13.74	7.28	12.76	12.99
	MB	0.79	9.14	12.36	6.57	12.22	12.55
	ANV	-1.94	6.79	12	5.99	10.4	11.8
	AV	-2.12	5.8	11.19	5.57	9.78	11.43
	Avg.	-0.357	7.841	12.101	5.783	11.245	12.019

F. Generalization Results

Table 10 presents the performance degradation values for both OrienterNet (Δ_{ONet}) and our proposed RHO model (Δ_{RHO}) when evaluated on the unseen city dataset. A comparative analysis of the average values demonstrates that RHO achieves significantly lower degradation values across both position and orientation metrics. For example, an average position recall degradation at 5m is 12.101 for Δ_{RHO} compared to 15.471 for Δ_{ONet} , and an average orientation recall degradation at 3° is 11.245 for Δ_{RHO} , compared to 16.052 for Δ_{ONet} . These results confirm that RHO maintains higher generalization capability and robustness in new environments.

G. More Qualitative Analysis

We show more qualitative results in Fig. 6, when the environment lacks distinctive elements, localizing a single image with OrienterNet often fails. Our proposed model RHO mitigates this by taking a whole panoramic view as input, which contains definitely richer elements than a single pinhole image. However, OSM is crowd-sourced, and the level of detail of the map is inconsistent and widely varies. When the elements do not appear in the map, or when lots of repeated elements exist, the pose estimation of our model could also be ambiguous.

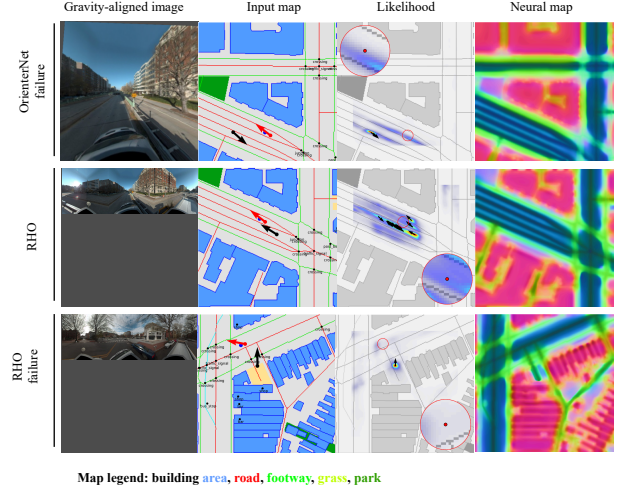


Figure 6. Failure cases of localization. Black arrow ($\rightarrow$) represents the pose estimation, and red arrow $\rightarrow$ indicates the ground truth.

H. Limitations and Future Work

While our model demonstrates remarkable performance on the OSM-based MCVGL task, our proposed SUM module and cross fusion logic are specifically designed for panoramic imagery processing. When applied to pinhole images alone, these mechanisms fail to boost performance compared to pinhole-branch architectures. This limitation restricts the model’s applicability to scenarios where only pinhole images are available. When training on the complete CV-RHO dataset containing over 2.7M images across all variations, our model achieves optimal checkpoints within early steps, despite iterative optimization of learning rates and optimizer configurations. This suggests that the current model architecture may not fully exploit the rich diversity of training samples.

Future work should focus on developing more sophisticated model architectures capable of processing multi-scale datasets and extending the two-branch framework to accommodate diverse input modalities. The evaluation across various datasets should be conducted to test our model’s generalization capability.

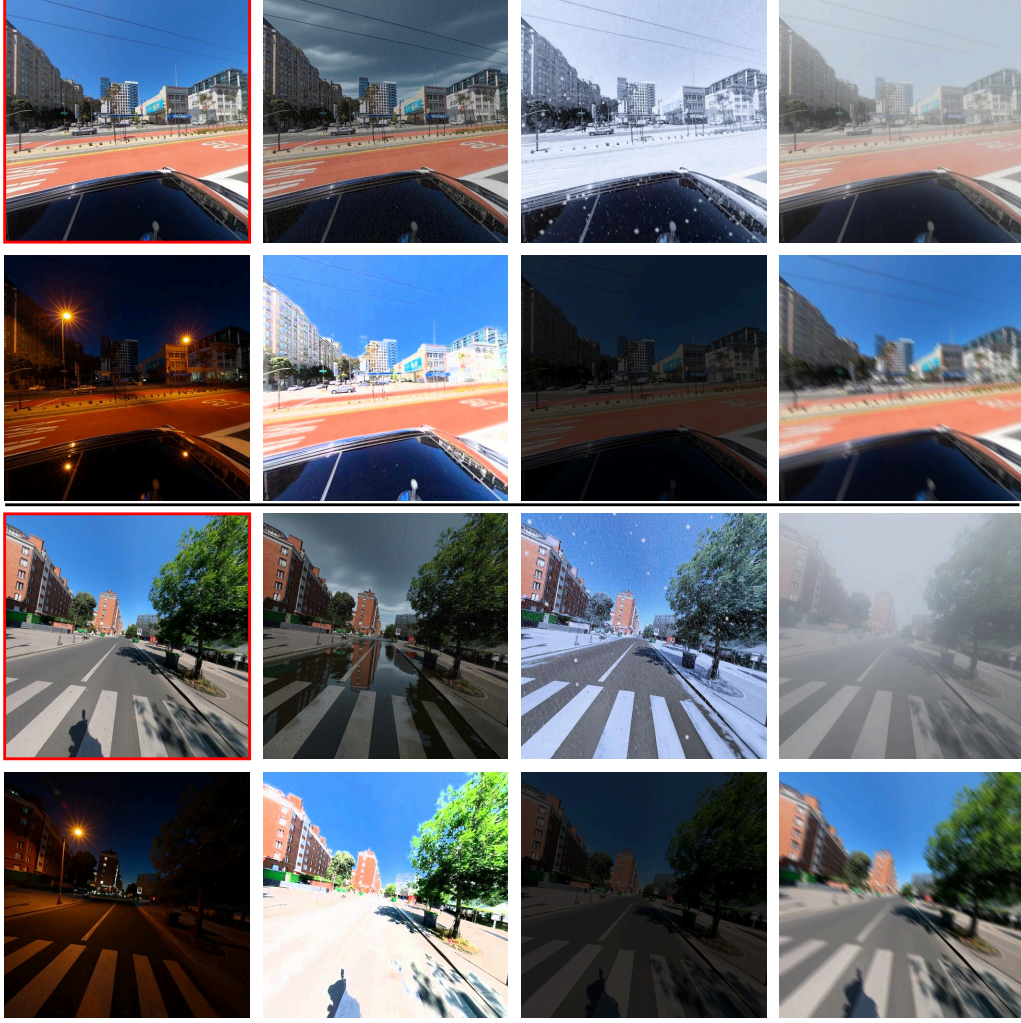
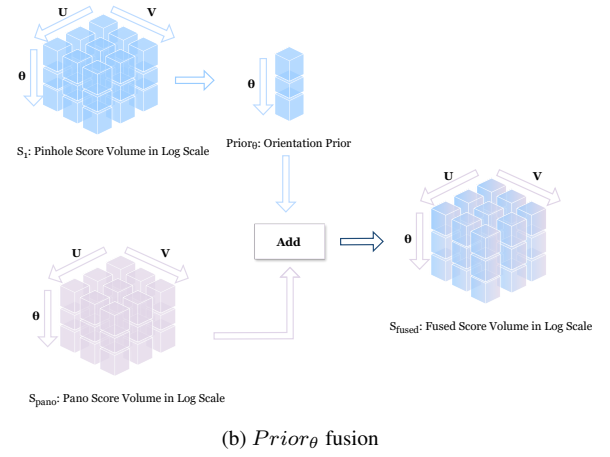
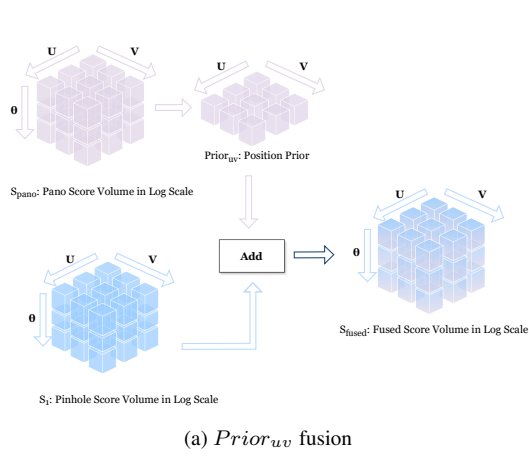


Figure 7. More samples of generated images. Images with red borders are original images.

Figure 8. One-way fusion strategy: (a) $Prior_{uv}$ fusion: fuse S_1 with position prior $Prior_{uv}$. (b) $Prior_{\theta}$ fusion: fuse S_{pano} with orientation prior $Prior_{\theta}$.

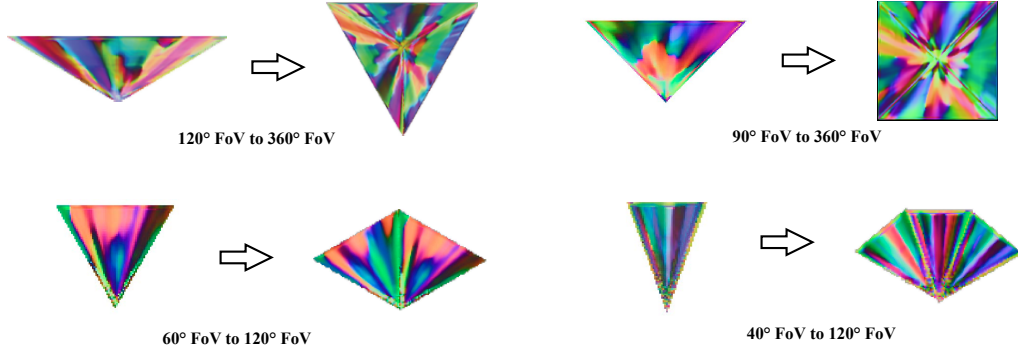


Figure 9. Different splits and BEV feature merge strategies.

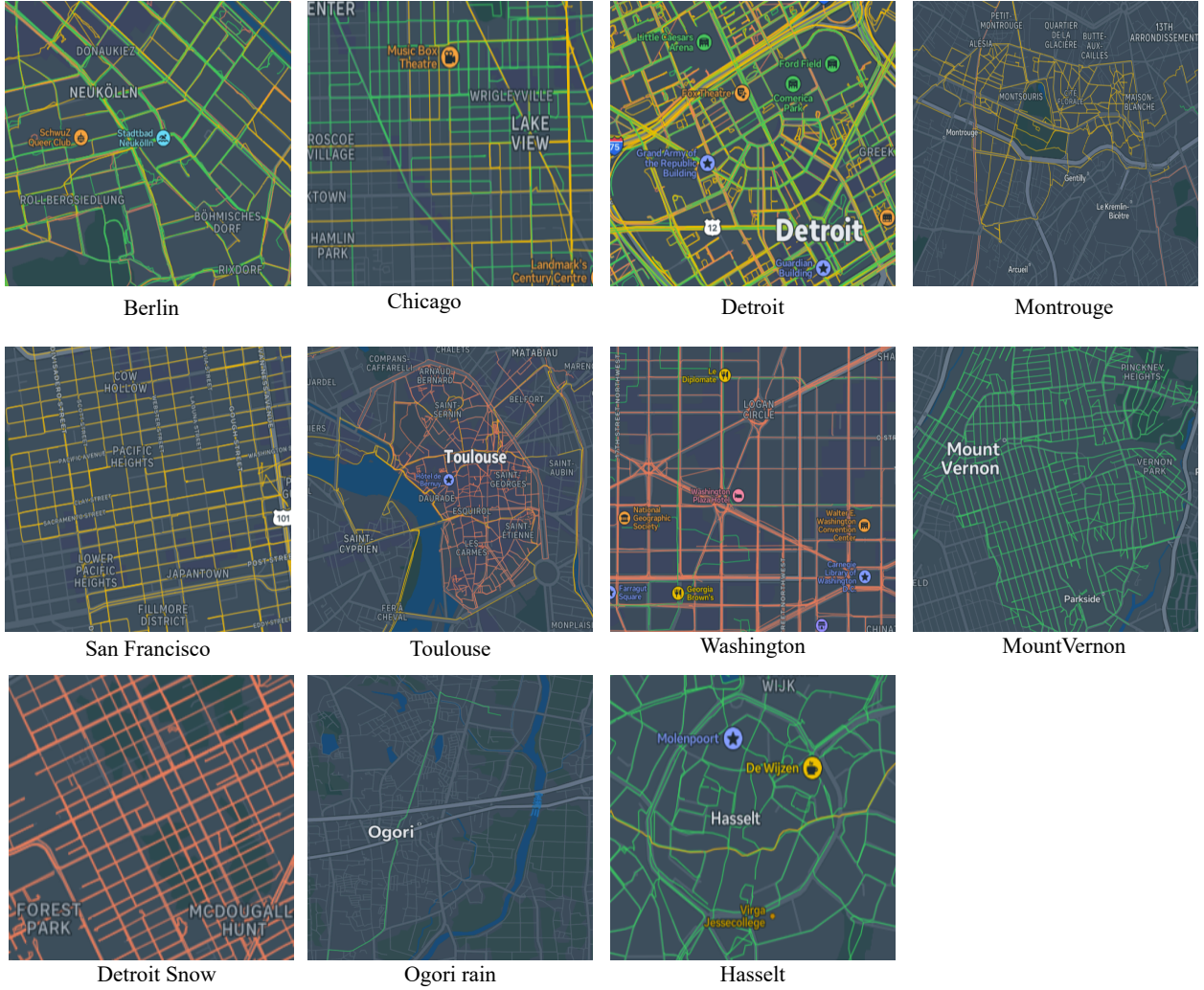


Figure 10. Selected data points of our CV-RHO dataset and Sim2Real across 11 cities. Color from red to green indicates capture date.